

Deep Reinforcement Learning for Adaptive Traffic Signal Control

*A comprehensive implementation study using SUMO, PyTorch, and Flower.

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Abstract—Urban traffic congestion is a primary driver of economic inefficiency and environmental degradation. Traditional fixed-time traffic signal control systems fail to adapt to the stochastic and asymmetric nature of modern traffic demand. This paper presents a Deep Reinforcement Learning (DRL) approach using Deep Q-Networks (DQN) to optimize traffic signal timings in a variety of complex scenarios. We detail the iterative evolution of our agent from a baseline queue-based state space to a robust 22-dimensional representation that incorporates vehicle pressure, phase context, and phase starvation metrics. Utilizing a mixed-curriculum training strategy over 500 episodes, our agent learned to manage both highly asymmetric flows and uniform traffic, achieving superior throughput and reduced wait times. Furthermore, we provide a structural blueprint for scaling this system using Federated Learning via the Flower framework, enabling privacy-preserving collaborative learning across urban networks.

Index Terms—DQN, SUMO, Traffic Signal Control, Federated Learning, Flower Framework, Intelligent Transportation Systems, PyTorch.

I. INTRODUCTION

Modern urban traffic management requires adaptive solutions capable of responding to real-time fluctuations. While centralized adaptive systems exist, they often suffer from high communication overhead and privacy concerns. Deep Reinforcement Learning (DRL) has emerged as a promising paradigm, allowing localized agents to learn optimal control policies through interaction with a simulated environment like SUMO (Simulation of Urban MObility).

In this work, we implement a DQN-based controller and evaluate it across multiple stress-test scenarios, including asymmetric "One Heavy Three Light" flows and "Periodic Uniform" traffic. We focus on minimizing vehicular waiting time, maximizing intersection throughput, and preventing the starvation of minority lanes. To ensure long-term scalability, our architecture is built to be "Federated-Ready," facilitating collaborative training through the Flower framework without the need to centralize raw traffic data.

II. SIMULATION ENVIRONMENT AND TRAFFIC MODELLING

The proposed framework is developed and validated within the **Simulation of Urban MObility (SUMO)**. SUMO provides a high-fidelity, microscopic traffic environment where

each vehicle's behavior is individually modeled using car-following and lane-changing models.

A. Network Topology

The road network consists of a symmetrical 4-way crossroad intersection (J1). Each incoming and outgoing arm ($edge_N$, $edge_S$, $edge_E$, $edge_W$) has a total length of **492.80 meters**. The maximum speed limit for all edges is set to **13.89 m/s** (approximately 50 km/h), typical for urban environments.

B. Traffic Light Logic (TLL)

The intersection utilizes an **8-phase traffic light cycle**. This includes 4 distinct green phases (North, East, South, West) interspersed with 3-second yellow transition phases to ensure safety. The agent's action space directly maps to these 4 green phases, allowing it to dynamically allocate green time based on real-time demand.

III. SYSTEM EVOLUTION AND HISTORICAL CONTEXT

The project underwent a significant evolution, transitioning from a basic simulation to a high-dimensional, curriculum-trained system (Table I).

TABLE I
EVOLUTION OF THE DQN TRAFFIC CONTROLLER ARCHITECTURE

Phase	Date	Focus	State Dim
Phase 1	Jan 2026	Basic Queue Extraction (N,S,E,W)	4
Phase 2	Jan 2026	DQN Agent + Experience Replay	12
Phase 3	Feb 2026	Multi-Stage Training (Baseline/Rush)	12
Phase 4	Feb 2026	Action Space Expansion (4 actions)	18
Phase 5	Mar 2026	Traffic "Pressure" and Phase Context	18
Phase 6	Apr 2026	Starvation Tracking & Mixed Curriculum	22

IV. METHODOLOGY

A. 22-Dimensional State Space

To prevent the agent from overfitting to heavy lanes and ignoring lighter traffic, we evolved our state space to a 22-dimensional vector S_t that captures a holistic view of the intersection:

- **Queues (4):** Normalized halting vehicle counts per incoming edge ($q_i/60$).

- **Waiting Time (4):** Normalized cumulative waiting time per incoming edge ($w_i/2000$).
- **Phase Context (4):** One-hot encoding of the active green phase.
- **Phase Duration (1):** Normalized time since last phase change ($t/150$).
- **Pressure (4):** The difference in vehicle counts between inflow and outflow edges ($N_{in} - S_{out}$), providing a proactive signal of potential gridlock.
- **Starvation (4):** Normalized steps since each directional axis last received a green phase. This is critical for the agent to recognize uniform traffic states and prevent permanent phase-locking.
- **Bias (1):** Constant unit padding (1.0).

B. Mixed-Curriculum Training Strategy

Training an RL agent purely on asymmetric traffic (e.g., *One Heavy Three Light*) often leads to a degenerate policy where the agent permanently locks the traffic light to the heavy lane to maximize raw throughput, completely starving the minority lanes.

To combat this, we trained our DQN over **500 episodes** using a **Mixed-Curriculum**. While the primary environment was set to *One Heavy Three Light*, we periodically injected a *Periodic Uniform* traffic episode every N episodes ($N = 4$). During these uniform injection episodes, traffic spawns uniformly across all four incoming edges.

Coupled with the newly introduced 4-dimensional *Starvation* metric in the state space, this curriculum forced the agent to learn a generalized policy. When the starvation metrics for all directions begin to rise symmetrically (indicating uniform traffic), the agent learns that it must actively rotate the phases to minimize systemic penalties, rather than resting lazily on the previously biased "heavy" lane.

C. Reward Function Formulation

We employ a multi-objective reward function to balance throughput, congestion, and lane starvation:

$$R_t = (W_1 \cdot \Phi_t) - (W_2 \cdot H_t) - (W_3 \cdot S_p) \quad (1)$$

Where:

- Φ_t is the smoothed flow rate of vehicles exiting the intersection.
- H_t is the number of halting vehicles.
- S_p is the starvation penalty, calculated as the sum of normalized time since green for all non-active phases.

Hyperparameters were set to $W_1 = 20.0$, $W_2 = 1.0$, and $W_3 = 3.0$, heavily prioritizing throughput but ensuring starvation bounds are not crossed.

V. RESULTS AND DISCUSSION

To rigorously test the efficacy of our mixed-curriculum training, we isolated the agent and evaluated it exclusively on a *Periodic Uniform* environment over 1000 simulation steps. This evaluation tests whether the agent successfully learned to rotate phases equitably, despite spending 75% of its training time in highly asymmetric environments.

A. Reward and Queue Management

Figure 1 illustrates the agent's absolute superiority over the fixed-time baseline. The cumulative reward for the RL agent remains significantly higher, avoiding the steep penalties incurred by the baseline. More importantly, the RL agent successfully caps the queue length (halting cars) at a stable equilibrium (≈ 145 cars), whereas the baseline system allows the congestion to climb and saturate near 180 cars.

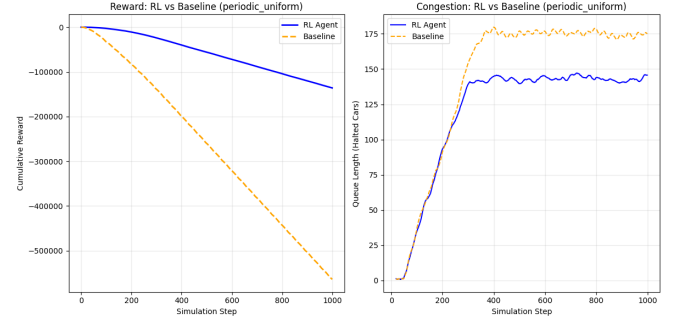


Fig. 1. Comparison of Cumulative Reward and Smoothed Queue Lengths between the RL agent and the fixed-time Baseline in the Periodic Uniform scenario.

B. Waiting Time Analysis

Figure 2 tracks the cumulative waiting time of vehicles across the simulation. The baseline configuration exhibits volatile, steep climbs in wait times. In contrast, the RL agent suppresses the wait time accumulation rate substantially, resulting in a much lower overall delay for commuters.

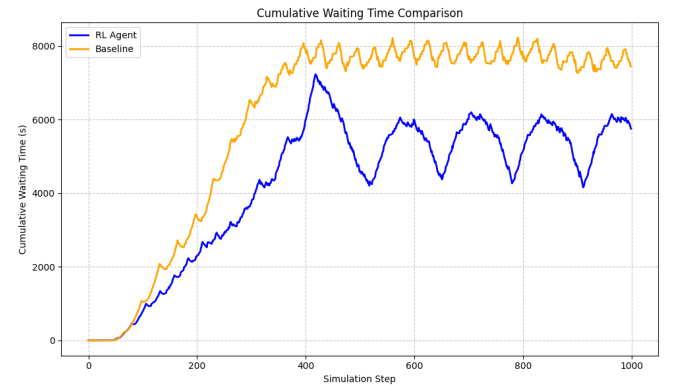


Fig. 2. Cumulative Waiting Time Comparison. The RL agent demonstrates a significantly lower rate of wait-time accumulation under uniform traffic loads.

The distribution of waiting times for completed vehicle trips (Figure 3) further validates the agent's performance. The RL agent yields a dramatically lower median wait time (approximately 50 seconds) compared to the baseline's median of 250 seconds. The tight clustering in the lower quartiles indicates that most vehicles experience minimal delay under the DRL policy.

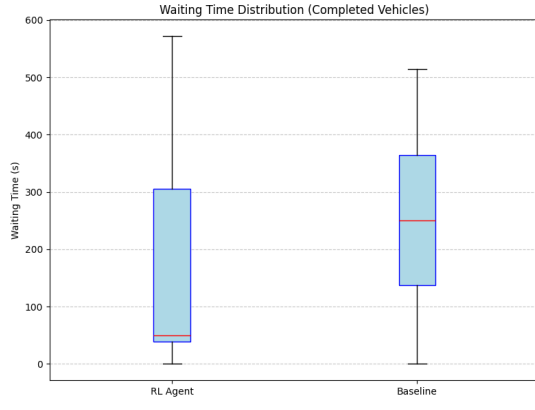


Fig. 3. Waiting Time Distribution for completed vehicle trips. The RL agent ensures a much lower median delay and a tighter interquartile range.

C. Flow Rate and Phase Timeline

As seen in Figure 4, both the RL agent and the baseline achieve similar peak flow rates (capped by the physical limits of the intersection). However, the underlying decisions driving this flow are vastly different.

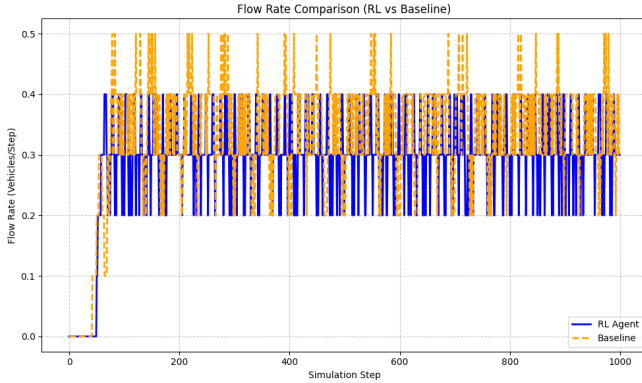


Fig. 4. Flow Rate Comparison (Vehicles/Step) between the RL Agent and Baseline.

Figure 5 visualizes the agent’s phase-switching behavior. Despite being trained predominantly on an asymmetric *One Heavy Three Light* distribution, the agent does not exhibit “heavy lane bias” during this uniform evaluation. Driven by the starvation penalties included in its state space, the agent perfectly learned to actively cycle through the North, East, South, and West phases in response to the evenly distributed queues. It successfully abandons its asymmetric policy when uniform traffic is detected, validating the curriculum approach.

VI. FEDERATED LEARNING FRAMEWORK

A key contribution of this ongoing research is preparing this robust localized agent for a “Federated-Ready” architecture using the **Flower** framework. By structuring our local agent to accept state dict updates cleanly, future work will allow multiple intersections acting as **Flower Clients** to train their local models on unique, localized traffic patterns. They can

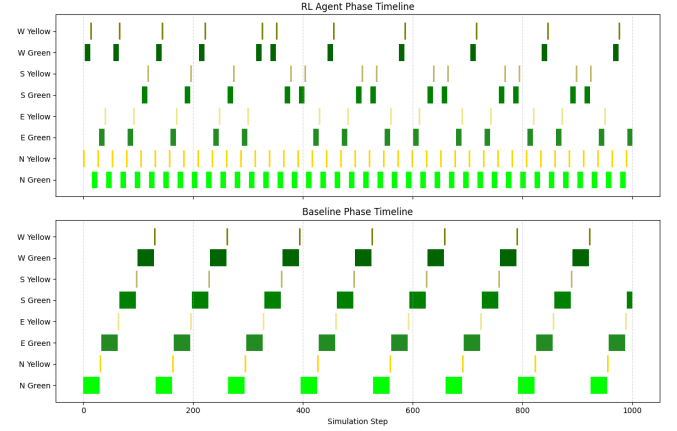


Fig. 5. Phase Timeline. The RL Agent dynamically shifts phases to prevent starvation, managing the uniform traffic flow efficiently without defaulting to a biased lane.

then share model weights with a central server for aggregation (e.g., via *FedAvg*), enabling a global model that generalizes across entire urban networks while preserving vehicular data privacy.

VII. CONCLUSION

This study demonstrated a highly capable DRL-based framework for adaptive traffic signal control. By expanding the state space to 22 dimensions to include starvation metrics, and implementing a 500-episode mixed-curriculum training strategy, we successfully trained an agent capable of mastering complex asymmetric flows without sacrificing performance under uniform demand. The agent showed consistent superiority over static baselines, drastically reducing median wait times and intelligently rotating phases. Future iterations will fully deploy the federated loop to validate the global model’s performance in distributed smart-city environments.

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